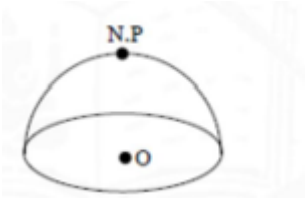
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

What is the potential difference between the north pole(NP) and the centre(O) of an inverted hemispherical conducting bowl of radius R carrying a uniform surface density ρ



(A) $\frac{\rho R}{2e_0}$

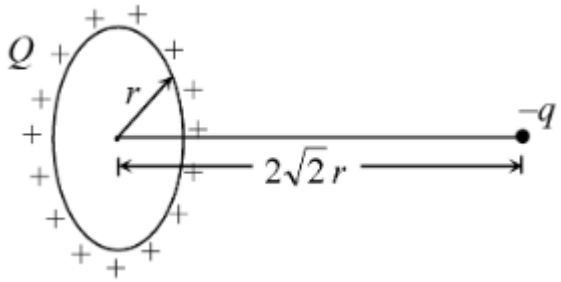
(B) $\frac{\rho R}{2e_0}(\sqrt{2}-1)$

(C) $\frac{\rho R^2}{2\epsilon_0}$

(D) $\sqrt{2}(\sqrt{2}-1)\frac{\rho R^2}{2\epsilon_0}$

Q.2

A charge $-q$ is placed at the axis of a charged ring of radius r at a distance of $2\sqrt{2}r$ as shown in figure. If ring is fixed and carrying a charge Q, the kinetic energy of charge $-q$ when it is released and reaches the centre of ring will be,



(A) $\frac{qQ}{4\pi\varepsilon_0 r}$

(B) $\frac{qQ}{12\pi\varepsilon_0 r}$

(C) $\frac{qQ}{6\pi\varepsilon_0 r}$

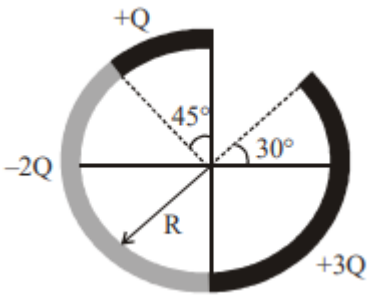
(D) $\frac{qQ}{2\pi\varepsilon_0 r}$

Q.3

Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the

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centre of curvature is -



(A) $\frac{Q}{2\pi\varepsilon_0 R}$

(B) $\frac{Q}{4\pi\varepsilon_0 R}$

(C) $\frac{2Q}{\pi\varepsilon_0 R}$

(D) $\frac{Q}{\pi\varepsilon_0 R}$

Q.4

Two point dipoles $p\hat{k}$ and $\frac{p}{2}\hat{k}$ are located at $(0, 0, 0)$ and $(1\ m, 0, 2\ m)$ respectively. The resultant electric field due to the two dipoles at the point $(1\ m, 0, 0)$ is -

(A) $\frac{9p}{32\pi\in_0}\hat{k}$

(B) $\frac{-7p}{32\pi\in_0}\hat{k}$

(C) $\frac{7p}{32\pi\in_0}\hat{k}$

(D) None of these

Q.5

Two infinite, parallel, non-conducting sheets carry equal positive charge density σ. One is placed in the yz plane and the other at distance $x = a$. Take potential $V = 0$ at $x = 0$. Choose the incorrect statements -

(A) For $0 \leq x \leq a$, potential $V_x = 0$

(B) For $x \geq a$, potential $V_x = -\frac{\sigma}{\epsilon_0}(x - a)$

(C) For $x \geq a$, potential $V_x = \frac{\sigma}{\epsilon_0}(x - a)$

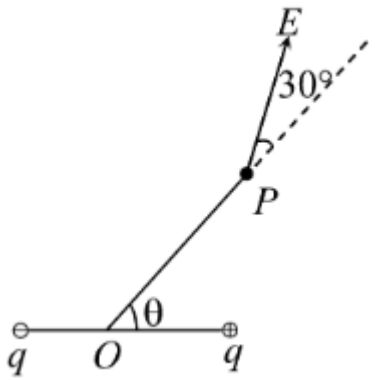
(D) For $x \leq 0$ potential $V_x = \frac{\sigma}{\epsilon_0}x$

Q.6

If net electric field E due to dipole at point P makes an angle 30° with the line OP as shown, then find the value

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of angle θ.



(A) $\tan^{-1}\left(\frac{1}{\sqrt{2}}\right)$

(B) $\tan^{-1}(\sqrt{2})$

(C) $\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

(D) $\tan^{-1}\left(\frac{\sqrt{3}}{2}\right)$

Q.7

A system consists of a uniform charged sphere of radius R and a surrounding medium filled by a charge with the volume density $\rho = \frac{\alpha}{\mathrm{r}}$, where α is a positive constant and r is the distance from the centre of the charge. The charge of the sphere for which the electric field intensity E outside the sphere is independent of r is -

(A) $\pi R^2\alpha$

(B) $4\pi R^2\alpha$

(C) $2\pi R^2\alpha$

(D) $3\pi R^2\alpha/4$

Q.8

Three identical positive charges Q are arranged at the vertices of an equilateral triangle. The side of the triangle is a. Find the intensity of the field at the vertex of a regular tetrahedron of which the triangle is the base.

(A) $\sqrt{6}\frac{\mathrm{KQ}}{\mathrm{a}^2}$

(B) $\sqrt{2}\frac{\mathrm{KQ}}{\mathrm{a}^2}$

(C) $\sqrt{3}\frac{\mathrm{KQ}}{\mathrm{a}^2}$

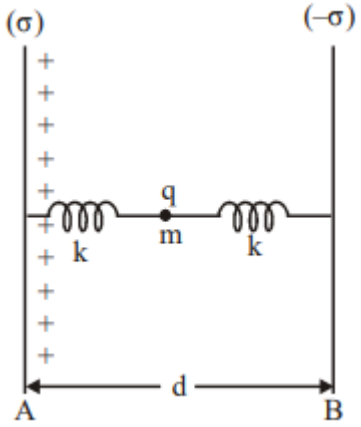
(D) None of these

Q.9

Two large non conducting plates having surface charge densities $+\sigma$ and $-\sigma$ respectively, are fixed d distance apart. A small test charge q of mass m is attached to two non conducting identical springs of spring constant k as shown in the adjacent figure. The charge q is now released from rest with springs in natural length. Then q will

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[neglect gravity]



(A) perform SHM with angular frequency $\sqrt{\frac{2k}{m}}$

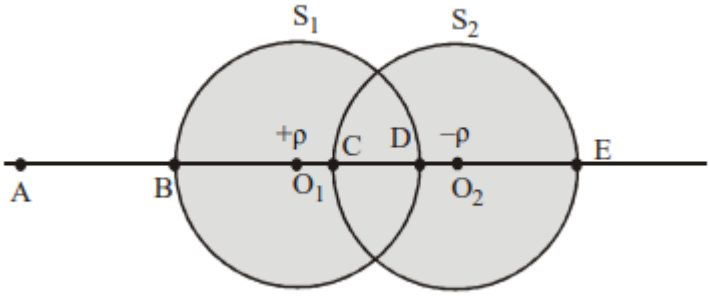
(B) perform SHM with amplitude $\frac{\sigma q}{2k\varepsilon_0}$

(C) not perform SHM, but will have a periodic motion

(D) remain stationary

Q.10

Two uniform spherical charge regions S_1 and S_2 having positive and negative charges overlap each other as shown in the figure. Point O_1 and O_2 are their centres and points A, B, C and D are on the lines joining centres O_1 and O_2. Choose the correct options -



(A) Electric field from A to B increases in magnitude

(B) Electric field from B to C first decreases then increases in magnitude

(C) Electric field from C to D remains constant in magnitude

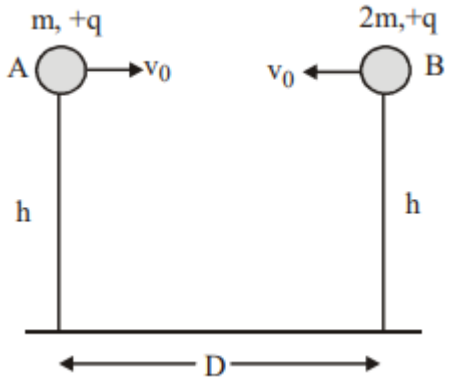
(D) Electric field from C to D increases in magnitude

Q.11

Two particles A and B have charge and mass $(+q, m)$ and $(+q, 2\ m)$ respectively. They are projected simultaneously from same height h horizontally towards each other with velocity v_0. Assume acceleration due to gravity to be uniform and vertically downward. Neglect the force due to magnetic field and effect of time varying magnetic and electric field. Assume that distance D is sufficiently large so that there is no collision between these

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particles. Select the correct statements -



(A) The time to hit the ground is independent of v_0.

(B) The two particles will be always at the same height

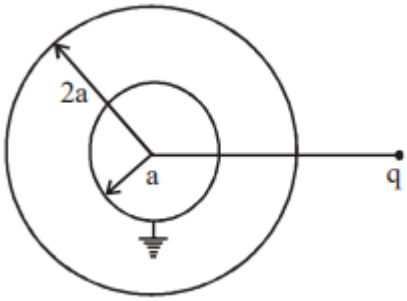
(C) The particle B will be always below the particle A.

(D) The particle A will be always below the particle B.

Q.12

Passage:

A solid conducting sphere of radius a is surrounded by a thin uncharged concentric conducting shell of radius $2a$. A point charge q is placed at a distance 4 a from common centre of conducting sphere and shell. The inner sphere is then grounded.



Question: The charge on solid sphere is -

(A) $-q/2$

(B) $-q/4$

(C) $-q/8$

(D) $-q/16$

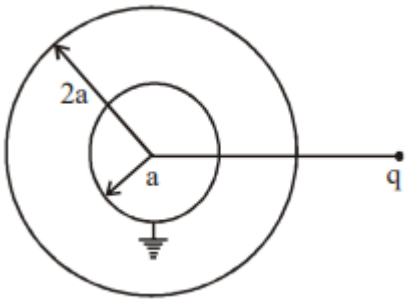
Q.13

Passage:

A solid conducting sphere of radius a is surrounded by a thin uncharged concentric conducting shell of radius $2a$. A point charge q is placed at a distance 4 a from common centre of conducting sphere and shell. The inner sphere is

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then grounded.



Question: Pick up the correct statement -

(A) Charge on surface of inner sphere is non-uniformly distributed.

(B) Charge on inner surface of outer shell is non-uniformly distributed.

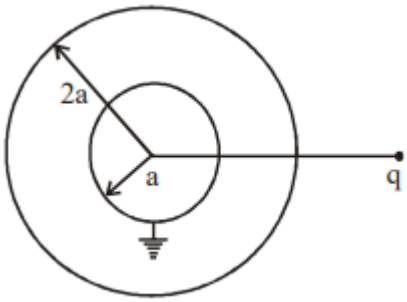
(C) Charge on outer surface of outer shell is non-uniformly distributed.

(D) All the above statements are false

Q.14

Passage:

A solid conducting sphere of radius a is surrounded by a thin uncharged concentric conducting shell of radius $2a$. A point charge q is placed at a distance 4 a from common centre of conducting sphere and shell. The inner sphere is then grounded.



Question: The potential of outer shell is -

(A) $\frac{q}{32\pi\varepsilon_0 a}$

(B) $\frac{q}{16\pi\varepsilon_0 a}$

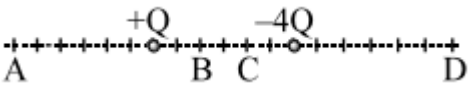
(C) $\frac{q}{8\pi\varepsilon_0 a}$

(D) $\frac{q}{4\pi\varepsilon_0 a}$

Q.15

Passage:

Related to the following diagram of two charges, $+Q$ and $-4Q$.



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Question: The net electric field is zero near which point?

(A) A

(B) B

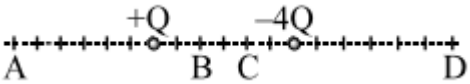
(C) C

(D) D

Q.16

Passage:

Related to the following diagram of two charges, $+Q$ and $-4Q$.



Question: At which point does the net electric field vector point to the left?

(A) A

(B) B

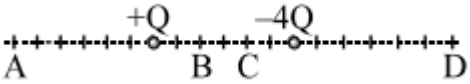
(C) C

(D) D

Q.17

Passage:

Related to the following diagram of two charges, $+Q$ and $-4Q$.



Question: Q.17 At which point would a small positive charge q feel the greatest force?

(A) A

(B) B

(C) C

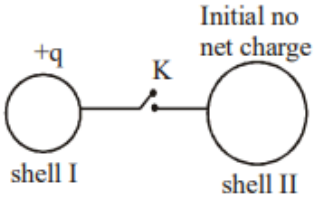
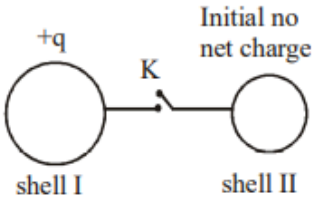
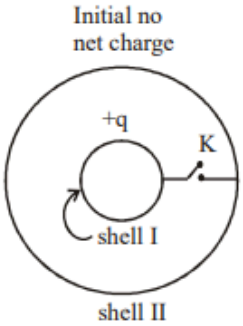
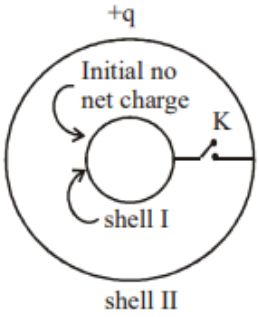
(D) D

Q.18

Column I gives certain situations involving two thin conducting shells connected by a conducting wire via a key K. In all situations one sphere has net charge $+q$ and other sphere has no net charge. After the key K is pressed,

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column II gives some resulting effect. Match the figures in column I with the statements in column II.

Column I	Column II
(A) 	(p) charge flows through connecting wire
(B) 	(q) Potential energy of system of spheres decreases.
(C) 	(r) No heat is produced
(D) 	(s) The sphere I has no charge after equilibrium is reached.

(A) A-(pq) B-(q) C-(pqs) D-(rs)

(B) A-(pq) B-(pq) C-(pqs) D-(rs)

(C) A-(pqs) B-(qr) C-(psq) D-(pq)

(D) A-(pqs) B-(pq) C-(pqs) D-(rp)

Q.19

A point charge q is placed at a distance a/2 directly above the centre of a horizontal square of side 'a' while another point charge $-q$ is placed at a distance ' a ' directly below one of its (the square's) corners. The electric flux through the square is $\frac{Aq}{24\varepsilon_0}$. Find the value of A.

Q.20

In a dark abandoned hall, there are a robber and a policeman standing against each other by the opposite sides of the hall. Having heard steps echoing in the darkness, the criminal points his gun horizontally on the unsuspecting cop and is about to shoot. What he doesn't know is that a potential difference $U = 144\ V$ was brought between the

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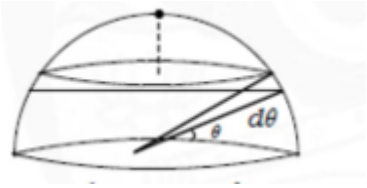
floor and the ceiling of the hall, thus creating a homogeneous field acting (vertically downwards) on the fired bullet, since the bullet acquires a charge q as it travels through the muzzle. Find the least charge-to-mass ratio q/m of the bullet such that it hits the floor before it reaches the policeman (m denote the mass of the bullet). The robber and the policemen are separated by a distance of $\mathrm{D} = 100\ \mathrm{m}$, the ceiling is $\mathrm{d} = 8\ \mathrm{m}$ above the floor and the bullet is fired horizontally with muzzle velocity $\mathrm{v}_0 = 400\ \mathrm{m\ s^{-1}}$. Assume that the robber holds his gun $\mathrm{h} = 2\ \mathrm{m}$ above the floor. $\left(\mathrm{g} = 10\ \mathrm{m\ s^{-2}}\right)$

Answers & Solutions

Q.1 Answer:

$\frac{\rho R}{2e_0}(\sqrt{2}-1)$

Solution:



Potential at center is half the potential of full spherical shell

$\therefore \quad \mathrm{Vcentre} = \frac{1}{2} \times \frac{Q}{4\pi\varepsilon OR} = \frac{6R}{2\epsilon 0}$

Potential at Pole,

V pole $= \frac{1}{4\pi\varepsilon_0}\int \frac{6dS}{r^3}$

$ds = 2\pi R\sin\theta R d\theta$

$r^3 = \sqrt{R^2+R^2-2R^2\cos\theta} = R\sqrt{2}\sqrt{1-\cos\theta}$

$\therefore V$ **pole** $= \frac{\sigma R}{2\sqrt{2}\varepsilon_0}\int_0^{\pi/2}\frac{\sin\theta d\theta}{\sqrt{1-\cos\theta}}$

$= \frac{\sigma R}{2\sqrt{2}\varepsilon\theta}\int_0^{\pi/2}\frac{d(1-\cos\theta)}{\sqrt{1-\cos\theta}}$

$= \frac{\sigma R}{2\sqrt{2}\varepsilon_0} 2\sqrt{1-\cos\theta}\Big|_0^{\pi/2}$

$= \frac{6R}{\sqrt{2}\varepsilon 0}$

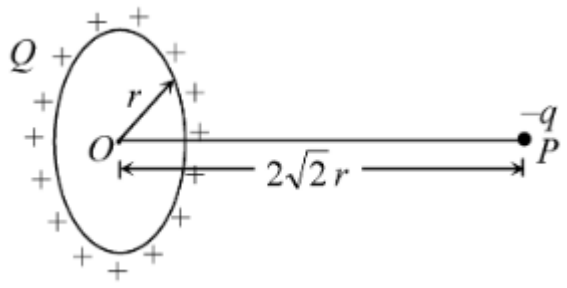
$\therefore V$ **diff** $= \Delta V = \frac{\sigma R}{2\varepsilon 0}(\sqrt{2}-1)$

Q.2 Answer:

$\frac{qQ}{6\pi\varepsilon_0 r}$

Solution:

K.E. of $-q$ **at** $O = q\,(V_O - V_P)$



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$$= q\left[\frac{Q}{4\pi\varepsilon_0 r} - \frac{Q}{4\pi\varepsilon_0\sqrt{r^2+(2\sqrt{2}r)^2}}\right]$$
$$= \frac{2}{3}\frac{qQ}{4\pi\varepsilon_0 r} = \frac{qQ}{6\pi\varepsilon_0 r}$$

Q.3 Answer:

$\frac{\mathrm{Q}}{2\pi\varepsilon_0 \mathrm{R}}$

Solution:

$$V = V_1 + V_2 + V_3 = \frac{1}{4\pi\epsilon_0}\cdot\frac{Q}{R} + \frac{1}{4\pi\epsilon_0}\left(\frac{-2Q}{R}\right) + \frac{1}{4\pi\epsilon_0}\left(\frac{3Q}{R}\right) = \frac{1}{4\pi\epsilon_0}\left(\frac{2Q}{R}\right)$$

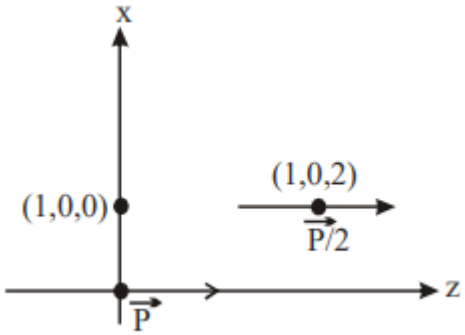
Q.4 Answer:

$\frac{-7\mathrm{p}}{32\pi\in_0}\hat{\mathrm{k}}$

Solution:

The given point is at axis of $\frac{\vec{\mathrm{P}}}{2}$ dipole and at equatorial line of $\vec{\mathrm{P}}$ dipole so that field at given point.

$\vec{\mathrm{E}} = -\frac{\mathrm{k}\vec{\mathrm{p}}}{(1)^3} + \frac{2\mathrm{k}(\vec{\mathrm{p}}/2)}{(2)^3} = \frac{-7\vec{\mathrm{p}}}{32\pi\varepsilon_0}$



Q.5 Answer:

For $x \geq a$, potential $V_x = \frac{\sigma}{\epsilon_0}(x-a)$

Solution:

$$0 \leq \mathrm{x} \leq \mathrm{a};\ \mathrm{V_x} = \left[-\int_0^{\mathrm{x}} \mathrm{E_x dx}\right] + \mathrm{V_{(0)}} = 0 \text{ as } (\mathrm{E_X} = 0)$$
$$\mathrm{x} \geq \mathrm{a};\ \mathrm{V_x} = \left[-\int_0^{\mathrm{x}} \mathrm{E_x dx}\right] + \mathrm{V_{(a)}} = \left[-\int_{\mathrm{a}}^{\mathrm{x}} \frac{\sigma}{\epsilon_0}\,\mathrm{dx}\right] + \mathrm{V_{(a)}} = -\frac{\sigma}{\epsilon_0}(\mathrm{x}-\mathrm{a})$$
$$\mathrm{x} \leq 0 \quad ;\mathrm{V_x} = -\int_0^{\mathrm{x}} \mathrm{E_x dx} + \mathrm{V_{(0)}} = -\left(-\frac{\sigma}{\epsilon_0}\mathrm{x}\right) + \mathrm{V_{(0)}} = \frac{\sigma}{\epsilon_0}\mathrm{x}$$

Q.6 Answer:

$\tan^{-1}\left(\frac{2}{\sqrt{3}}\right)$

Solution:

$$\tan 30^\circ = \frac{1}{2}\tan\theta, \quad \tan\theta = \frac{2}{\sqrt{3}}$$

Q.7 Answer:

$2\pi R^2\alpha$

Solution:

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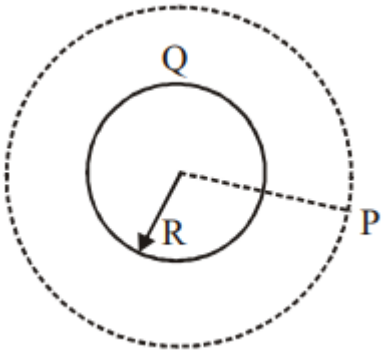
$\oint \mathrm{E_p dS} = \frac{\mathrm{Q+Q'}}{\varepsilon_0}$, where $\mathrm{Q'}$ is the charge outside the sphere

$\mathrm{Q_1' = \int_R^r dV = \int_R^r \frac{\alpha}{r} \times 4\pi r^2 dr = 4\pi\alpha \left(\frac{r^2}{2}\right)_R^r = 4\pi\alpha \left(\frac{r^2}{2} - \frac{R^2}{2}\right) = 2\pi\alpha \left(r^2 - R^2\right)}$

$E_p \times 4\pi r^2 = \frac{Q+2\pi\alpha\left(r^2-R^2\right)}{\varepsilon_0}$

$E_p = \frac{Q}{4\pi r^2 \varepsilon_0} + \frac{\alpha}{2\varepsilon_0} - \frac{\alpha R^2}{2r^2\varepsilon_0}$

E is independent of r if $\frac{Q}{4\pi r^2 \varepsilon_0} - \frac{\alpha R^2}{2r^2\varepsilon_0} = 0$



Q.8 Answer:

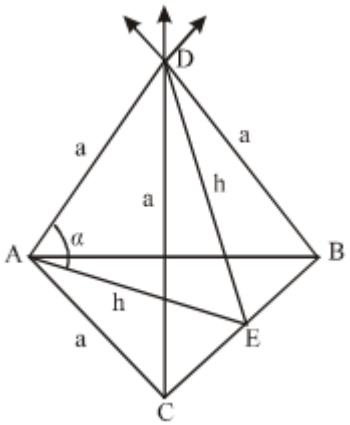
$\sqrt{6}\frac{\mathrm{KQ}}{\mathrm{a^2}}$

Solution:

Each charge creates at point D a field intensity of $\mathrm{E_1 = \frac{KQ}{a^2}}$. The total intensity will be the sum of three vectors (fig.). The sum of the horizontal components of these vectors will be zero, since they are equal in magnitude and form angles of 120° with each other. The vectors from angles of $90^\circ -$ a with the vertical, where a is the angle between the edge of the tetrahedron and the altitude h of triangle ABC.

The vertical components are identical each being equal to $\frac{\mathrm{KQ}}{\mathrm{a^2}} \sin\alpha$. It follows from triangle ADE that $\sin\alpha = \sqrt{\frac{2}{3}}$.

Therefore, the sought intensity of the field is $E = \sqrt{6}\frac{KQ}{a^2}$.



Q.9 Answer:

perform SHM with angular frequency $\sqrt{\frac{2\mathrm{k}}{\mathrm{m}}}$

Solution:

Equilibrium position $2\mathrm{kx_0} = \frac{\sigma \mathrm{q}}{\varepsilon_0} \Rightarrow \mathrm{x_0} = \frac{\sigma \mathrm{q}}{2\mathrm{k}\varepsilon_0}$

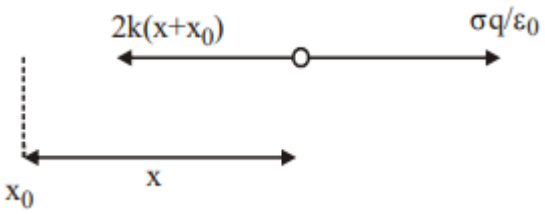
$\mathrm{x_0}$ is measured from position where springs were in natural length.

For (A) At a position x right of the equilibrium

$m\ddot{\mathrm{x}} = -2\mathrm{k}\left(\mathrm{x} + \mathrm{x_0}\right) + \frac{\sigma \mathrm{q}}{\varepsilon_0} \Rightarrow m\ddot{\mathrm{x}} = -2\mathrm{kx}$

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Angular frequency $\omega = \sqrt{\frac{2k}{m}}$, **Amplitude of SHM** $x_0 = \frac{\sigma q}{2k\varepsilon_0}$

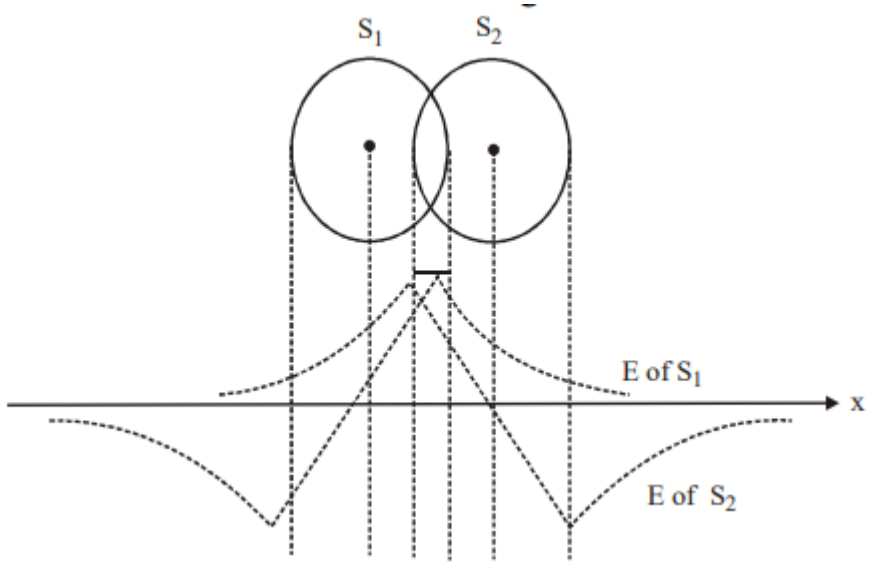


Q.10 Answer:

Electric field from A to B increases in magnitude

Solution:

Using superposition we can now estimate magnitude and direction of net electric field.



Q.11 Answer:

The time to hit the ground is independent of v_0.

Solution:

Due to gravitational force, the two particles will fall same height in same time H. Hence the electrostatic force will not have component in vertical direction, hence the time to hit the ground will be $\sqrt{\frac{2h}{g}}$, **independent of** v_0.

Q.12 Answer:

$-q/4$

Solution:

The inner sphere is grounded, hence its potential, hence its potential is zero.

The net charge on isolated outer sphere is zero.

Let the charge on inner sphere be q'.

∴ Potential at centre of inner sphere is

$$\frac{1}{4\pi\varepsilon_0}\frac{q'}{a} + 0 + \frac{1}{4\pi\varepsilon_0}\frac{q}{4a} = 0 \quad \therefore q' = -\frac{q}{4}$$

The region in between conducting sphere and shell is shielded from charges on and outside the outer surface of shell. Hence charge distribution on surface of sphere and inner surface of shell is uniform. The distribution of induced charge on outer surface of shell depends only on point charge q, hence is non-

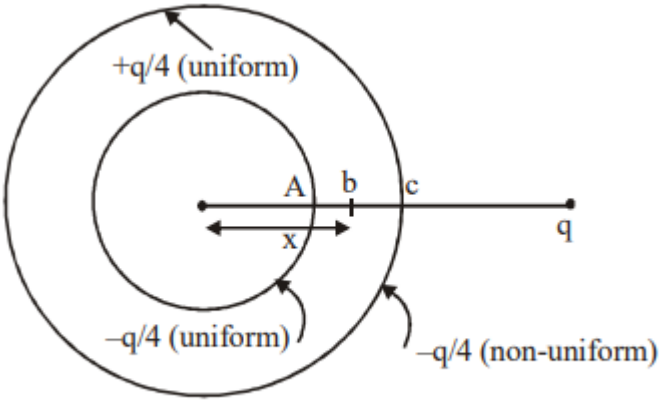
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uniform.

The charge distribution on all surfaces, is as shown.

The electric field at B is $= \frac{1}{4\pi\varepsilon_0}\frac{q}{4x^2}$ **towards left**

$\therefore V_C = V_C - V_A = -\int_{2a}^{a}\frac{1}{4\pi\varepsilon_0}\frac{q}{4x^2}dx = \frac{1}{32\pi\varepsilon_0}\frac{q}{a}$



Q.13 Answer:

Charge on outer surface of outer shell is non-uniformly distributed.

Solution:

The inner sphere is grounded, hence its potential, hence its potential is zero.

The net charge on isolated outer sphere is zero.

Let the charge on inner sphere be q'.

∴ Potential at centre of inner sphere is

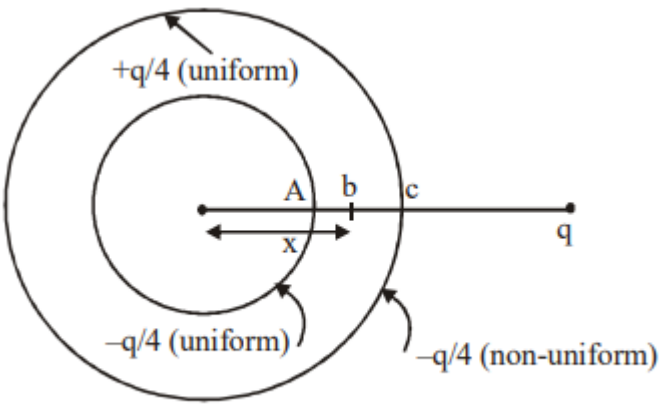
$\frac{1}{4\pi\varepsilon_0}\frac{q'}{a} + 0 + \frac{1}{4\pi\varepsilon_0}\frac{q}{4a} = 0 \quad \therefore q' = -\frac{q}{4}$

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The charge distribution on all surfaces, is as shown.

The electric field at B is $= \frac{1}{4\pi\varepsilon_0}\frac{q}{4x^2}$ **towards left**

$\therefore V_C = V_C - V_A = -\int_{2a}^{a}\frac{1}{4\pi\varepsilon_0}\frac{q}{4x^2}dx = \frac{1}{32\pi\varepsilon_0}\frac{q}{a}$



Q.14 Answer:

$\frac{q}{32\pi\varepsilon_0 a}$

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Solution:

The inner sphere is grounded, hence its potential, hence its potential is zero.

The net charge on isolated outer sphere is zero.

Let the charge on inner sphere be q'.

∴ Potential at centre of inner sphere is

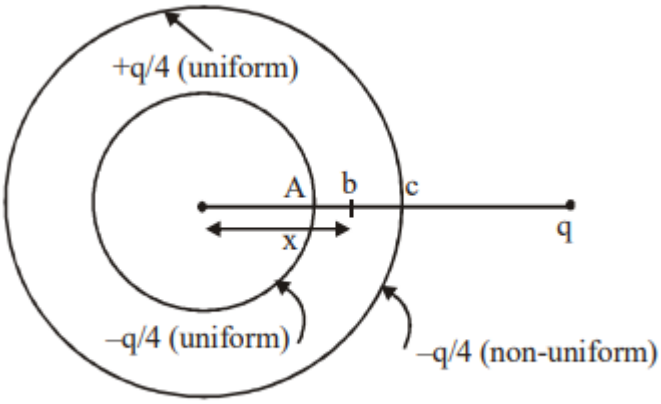
$\frac{1}{4\pi\varepsilon_0}\frac{q'}{a} + 0 + \frac{1}{4\pi\varepsilon_0}\frac{q}{4a} = 0 \quad \therefore q' = -\frac{q}{4}$

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Q.15 Answer:

A

Solution:

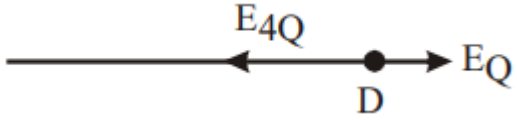
Electric field is zero nearer to smaller charge, not between charges (for opposite charges)

Q.16 Answer:

D

Solution:

At point D



Q.17 Answer:

C

Solution:

For max. force, electric field should be maximum i.e. possible at C.

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Q.18 Answer:

A-(pq) B-(pq) C-(pqs) D-(rs)

Solution:

$(A)-pq,(B)-pq,(C)-pqs,(D)-rs$

In situation A, B, and C, shells I and II are not at same potential. Hence charge shall flow from sphere I to sphere II till both acquire same potential.

If charge flows, the potential energy of system decreases and heat is produced.

If situations A and B charges shall divide in some fixed ratio, but in situation C complete charge shall be transferred to shell II for potential of shell I and II to be same.

In situation D both the shells are at same potential, hence no charge flows through connecting wire.

Q.19 Answer:

—

Solution:

Flux due to $+\mathrm{q}$ is $\phi_1=\frac{\mathrm{q}}{6\varepsilon_0}$

Flux due to $-\mathrm{q}$ is $\phi_2=\frac{1}{3}\left(\frac{\mathrm{q}}{8\varepsilon_0}\right)=\frac{\mathrm{q}}{24\varepsilon_0}$

(q can be enclosed by 8 imaginary cubes, flux through 3 faces (E $\perp$ A) will be zero) Flux due to $-\mathrm{q}$ is also in same direction because it is kept below the square.

$\phi=\phi_1+\phi_2=\frac{5\mathrm{q}}{24\varepsilon_0}$

Q.20 Answer:

—

Solution:

$$\mathrm{E}=\mathrm{U}/\mathrm{d}$$

$$\mathrm{a}=\mathrm{g}'=\frac{\mathrm{F_g}+\mathrm{F}_e}{\mathrm{m}}=\mathrm{g}+\frac{\mathrm{Uq}}{\mathrm{md}}$$

$$\Rightarrow \quad \dot{\mathrm{D}}=\sqrt{\frac{2\,\mathrm{h}}{\mathrm{g}'}}\mathrm{v}_0$$

$$\Rightarrow \mathrm{D}=\sqrt{\frac{2\,\mathrm{h}}{\mathrm{g}+\frac{\mathrm{Uq}}{\mathrm{md}}}}\mathrm{v}_0$$

On solving $\frac{\mathrm{q}}{\mathrm{m}}=3\mathrm{Ckg}^{-1}$.

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